

☆ Pipes & Cistern

Empty (-)
Filling (+)

$$A \& B = \frac{1}{6} + \frac{1}{5} = \frac{11}{30}$$

$$\frac{30}{11} = 2\frac{8}{11} \text{ hrs.}$$

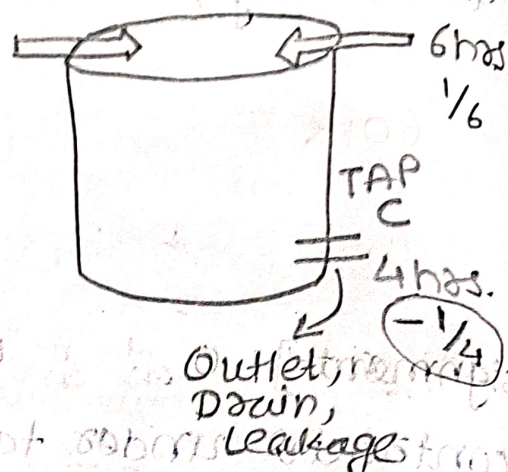
$$A \& B \& C = \frac{1}{6} + \frac{1}{5} - \frac{1}{4}$$

$$= \frac{22-15}{60} = \frac{7}{60}$$

$$\frac{60}{7} \Rightarrow 8\frac{4}{7} \text{ hrs.}$$

If number is (+ve) then tank must be filled

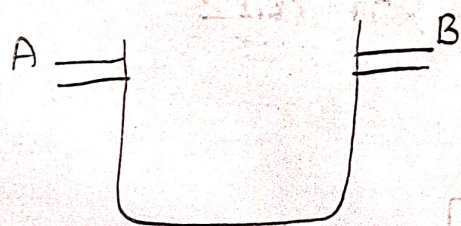
If number is (-ve) then tank must be emptied



Que Two pipes A and B can fill a tank in 8 mins & 12 mins respectively. Both the pipes are opened together and after 3 minutes, pipe A is turned off. What is total time required to fill the tank?

- (a) 3 minutes (b) 7.5 min (c) 10 min (d) 4.5 min

Sol
A $\Rightarrow$ 8 min $\Rightarrow \frac{1}{8}$
B $\Rightarrow$ 12 min $\Rightarrow \frac{1}{12}$



B is continuously working

Total time $\Rightarrow x$
Work done by B = $\frac{x}{12}$

$$\frac{x}{12} + \left(\frac{3}{8}\right) = 1$$

$\frac{1}{8}$ in 8 min $\Rightarrow \frac{3}{8}$
For 3 min $\Rightarrow \frac{3}{8}$

$2x + 3 = 12$
 $2x = 9$
 $x = 4.5 \text{ min}$

Que Pipes A and B can fill tanks in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long it will take to fill the tank? [Assignment]

(a) 4.5 hrs (b) 2 hrs (c) 6 hrs (d) 3.2 hrs

Sol

$$\begin{aligned}
 A &= 1/4 \\
 B &= 1/8 \\
 C &= -1/16 \\
 A+B+C &= \frac{1}{4} + \frac{1}{8} - \frac{1}{16} \\
 &= \frac{4}{16} + \frac{2}{16} - \frac{1}{16} \\
 &= \frac{4+2-1}{16} \\
 &= \frac{5}{16} \Rightarrow 16/5 \text{ hrs} \\
 &\Rightarrow 3.2 \text{ hrs}
 \end{aligned}$$

Que A tank can be filled by 20 buckets each capacity 13.5 litres. The capacity of each bucket be 9 litres, how many buckets will fill the same tank. [Assignment]

(a) 30 (b) 25 (c) 20 (d) 15.

Sol

$$\begin{aligned}
 \text{Tank capacity} &= 20 \times 13.5 \\
 &= 270 \text{ litres} \\
 270/9 &= \underline{\underline{30 \text{ buckets}}}
 \end{aligned}$$

OR CHAIN RULE

	No. of Bucket	Cap.
	20	13.5
more less	↑ ?	↓ 9

$$\begin{aligned}
 &\Rightarrow 20 \times \frac{13.5}{9} \\
 &\Rightarrow \frac{270}{9} \Rightarrow \underline{\underline{30 \text{ litres}}}
 \end{aligned}$$

Pipes & Cistern.
 $\Rightarrow$ Time & Work

Que

A tank consist of 3 taps A, B and C. Tap A & Tap B can fill the tank in 20 hrs and 12 hrs respectively, whereas tap C can empty the tank in 18 hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A.... so on. In how many hours the tank would be filled?

Sol

$$\begin{aligned} A &\Rightarrow 20 \text{ hrs} \Rightarrow \frac{1}{20} \\ B &\Rightarrow 12 \text{ hrs} \Rightarrow \frac{1}{12} \\ C &\Rightarrow 18 \text{ hrs} \Rightarrow \frac{1}{18} \end{aligned} \left. \begin{array}{l} \text{filling} \\ \text{empty} \end{array} \right\}$$

1	2	3	4	5
A	B	A	B	A
8	8	8	8	8
C	C	C	C	C

1st hour $\Rightarrow \frac{1}{20} - \frac{1}{18} \Rightarrow \frac{9-10}{180} \Rightarrow -\frac{1}{180}$ } In 1st hour Tank remains empty only.

2nd & 3rd hour $\Rightarrow \frac{1}{12} + \frac{1}{18} \Rightarrow \frac{3+2}{18} \Rightarrow \frac{5}{18}$ } Tank starts filling....

3rd hour $\Rightarrow -\frac{1}{180}$ } $\therefore$ 2nd & 3rd hour $\Rightarrow \frac{5}{180} - \frac{1}{180} \Rightarrow \frac{4}{180}$

$\therefore \frac{4}{180}$ into 2 hours... $\therefore \frac{4}{180} = 2 \text{ hrs}$ (*45 both side)

$\Rightarrow \frac{4}{180} = 90 \text{ hrs} + 2 \text{ hrs} \Rightarrow 92 \text{ hrs}$ X

$\Rightarrow 92 \text{ hrs}$ is wrong because at some time after 88 hrs there is some time where the tank must be full...

$\Rightarrow 92 \text{ hrs}$ is odd $\therefore$ A & B are working.

$\Rightarrow$ They remove water. How is possible that we are removing water and tank being full.

$\therefore \frac{4}{180} = 2$ (*44 Both side)
 $\Rightarrow \frac{176}{180} = 88 \text{ hrs}$

88 hrs + 2 hrs (for full tank)
 $\Rightarrow \underline{90 \text{ hrs}}$

89 hours $\rightarrow$ odd

Total filled $\Rightarrow \frac{44}{45}$

Remaining $\Rightarrow 1 - \frac{44}{45}$
 $\Rightarrow \frac{1}{45}$

$\therefore$ Total hours completed
 $\Rightarrow 89$ hrs

Extra $\frac{4}{5}$ hrs done by
BSC.

$\therefore$ Ans $\Rightarrow 89 \frac{4}{5}$ hrs

$\therefore$ 89 hours $\rightarrow$ odd $\rightarrow$ Next

90 hrs $\rightarrow$ even.

$\therefore$ BSC work.

$\Rightarrow$ BSC $\Rightarrow \frac{5}{180} \Rightarrow \frac{1}{36}$

$\therefore$ Total $\Rightarrow 36$ hrs.

$\therefore 1 \frac{4}{5} = 36$

$\leftarrow \frac{1}{45} = 20\%$

$\Rightarrow \frac{1}{45} \times 36 \times 4 \Rightarrow \frac{4}{5}$

one
If you want to know more about the speed of the journey, please go to the website of the journey.